

INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

Pós-Graduação em Sistemas Computacionais Embebidos
Sistemas Operativos de Tempo-Real (PGSCE-SIOPTR)

SIOPTR: TT4: Week 3

My Own KERnel (MOKER)

version: Setup

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2025/26

1 Tools and packages

In order to install the required tools and packages:

1. Open a terminal
2. Type the following Linux commands:
> `sudo apt-get update`
3. Then, install the required tools
> `sudo apt-get install libssl-dev`
> `sudo apt-get install libncurses5-dev`
> `sudo apt-get install libelf-dev`
> `sudo apt-get install bison`
> `sudo apt-get install flex`

or

```
> sudo apt-get install libssl-dev libncurses5-dev libelf-dev  
bison flex
```

4. Finally, reboot the system typing:
> `sudo reboot`

2 Showing grub menu

It is a good practice, to show the grub menu during the system boot. Because this allows us to choose the kernel version to boot.

For that purpose, we have to change the `/etc/default/grub` grub configuration file to show the grub menu.

1. So, for that type:
`sudo gedit /etc/default/grub`

```

1  # If you change this file, run 'update-grub' afterwards to update
2  # /boot/grub/grub.cfg.
3  # For full documentation of the options in this file, see:
4  # info -f grub -n 'Simple configuration'
5
6  GRUB_DEFAULT=0
7  GRUB_TIMEOUT_STYLE=hidden
8  GRUB_TIMEOUT=0
9  GRUB_DISTRIBUTOR='lsb_release -i -s 2> /dev/null || echo Debian'
10 GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
11 GRUB_CMDLINE_LINUX=""
12
13 ...

```

2. Comment line 7 with # and change from 0 to 10 the GRUB_TIMEOUT (line 8) option (the updated lines are blue colored):

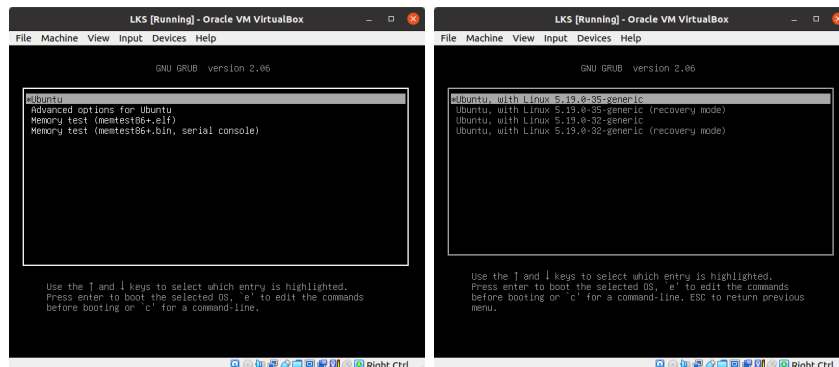
```

1  # If you change this file, run 'update-grub' afterwards to update
2  # /boot/grub/grub.cfg.
3  # For full documentation of the options in this file, see:
4  # info -f grub -n 'Simple configuration'
5
6  GRUB_DEFAULT=0
7  # GRUB_TIMEOUT_STYLE=hidden
8  GRUB_TIMEOUT= 10
9  GRUB_DISTRIBUTOR='lsb_release -i -s 2> /dev/null || echo Debian'
10 GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
11 GRUB_CMDLINE_LINUX=""
12
13 ...

```

3. Save and close
4. Next step is to update the grub. For that, type:
sudo update-grub2
5. Finally, reboot the system:
sudo reboot

At boot process, the grub menu is showed, use the "up" and "down" arrow keys and the "Enter" key to select an menu option. You have 10 seconds for that, otherwise the default kernel version is booted. If you select "Advanced options for Ubuntu" and press "Enter" key, then using "up" and "down" keys you are able to select the kernel version you want to boot.



3 Get Linux kernel source code

Now, it is time to create our own kernel version.

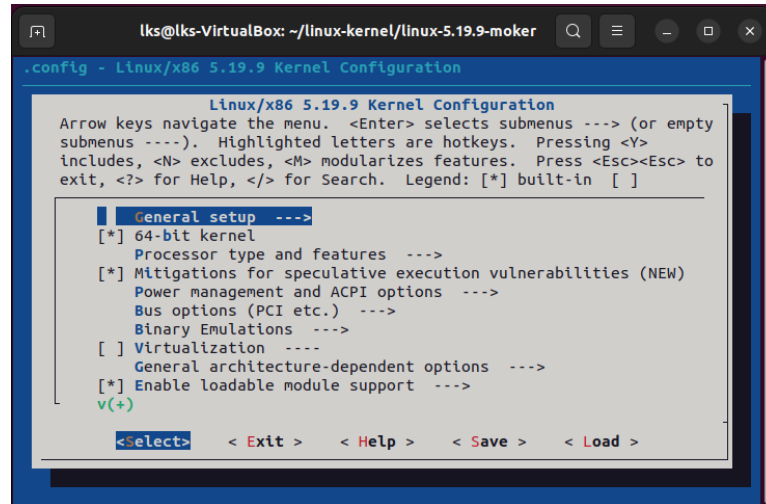
1. For that, create the `$HOME/linux-kernel` directory:
`mkdir linux-kernel`
2. Change to that directory:
`cd linux-kernel` Next get the source code from <http://kernel.org>
`./`. For that type:
`wget https://www.kernel.org/pub/linux/kernel/v6.x/linux-6.19.9.tar.gz`
3. Extract tar (.tar.gz) file:
`tar -xzf linux-6.19.9.tar.gz`
4. Rename linux-6.19.9 directory to linux-6.19.9-moker typing:
`mv linux-6.19.9 linux-6.19.9-moker`

4 Configuring Linux kernel

The next step is to configure the Linux kernel. In this case, we will not change nothing. However, we have to performing the following steps:

1. Open a terminal, and go to the `$HOME/linux-kernel` directory:
`cd linux-kernel`
`cd linux-6.19.9-moker`
2. Create the `.config` file by using the default symbol values
`make ARCH=x86 defconfig`

3. In order to customize the build and kernel options type:
`make menuconfig`



4. Using "Up", "Down", "Left" and "Right", "Space" and "Enter" keys you could select the appropriate compiling options. In this case, choose the "Exit".
5. Return to the `linux-kernel` directory, by typing:
`cd ..`

5 Customize Linux kernel version

In order to customize the Linux kernel version, edit `Makefile` file by adding `-moker` to `EXTRAVERSION` option.

1. Open a terminal, and go to the `$HOME/linux-kernel` directory:
2. For that, type:
`gedit linux-6.19.9-moker/Makefile`

cd li

```
1  # SPDX-License-Identifier: GPL-2.0
2  VERSION = 6
3  PATCHLEVEL = 19
4  SUBLEVEL = 9
5  EXTRAVERSION = -moker
6  NAME = Baby Opossum Posse
7
8  # *DOCUMENTATION*
9  # To see a list of typical targets execute "make help"
10 # More info can be located in ./README
11 # Comments in this file are targeted only to the developer, do not
12 # expect to learn how to build the kernel reading this file.
13 ...
```

3. Save and close.

6 Compiling and installing Linux kernel

For compiling and installing the Linux kernel, it is necessary to execute a set of commands. The first step is to open a terminal, and go to the \$HOME/linux-kernel directory.

Then, perform a:

- Sequence of commands:
cd linux-6.19.9-moker
make
sudo make modules_install
sudo make install
cd ..
sudo update-grub2
sudo reboot
- Alternatively (preferably), you can :
 1. Download the kcompile.sh script file from https://drive.google.com/file/d/1S6Inj_eDavsCokNaNz-2EjXBMSm7HLG4/view?usp=sharing.

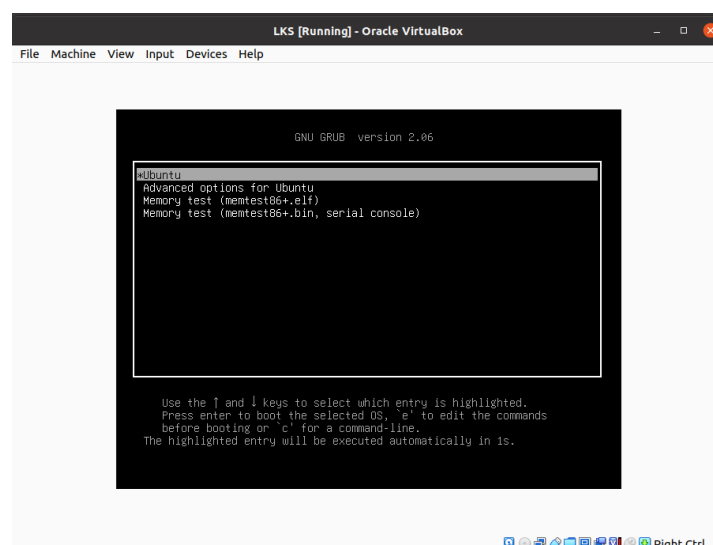
```
1  #!/bin/bash
2  start=$(date +%s')
3  cd linux-6.19.9-moker
4  sudo make 2>./errors-6.19.9-moker
5  sudo make modules_install
6  sudo make install
7  cd ..
8  sudo update-grub2
9  cat errors-6.19.9-moker
10 echo "Linux kernel compilation and installation took $((date +%s') - $start)) seconds"
```

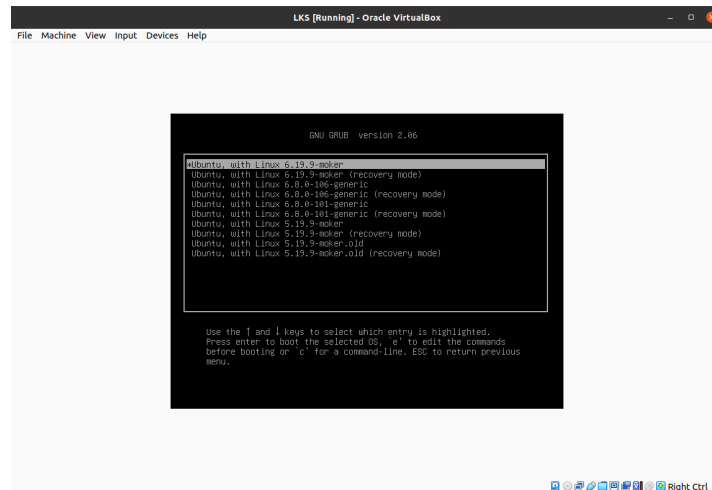
2. Assuming that `kcompile.sh` script file is into `Downloads` directory. Copy the `kcompile.sh` script file to `linux-kernel` directory:
`cp ~/Downloads/kcompile.sh .`
3. Assign execution permission to the `kcompile.sh` script file, by:
`sudo chmod 755 kcompile.sh`
4. The next step, is to execute `kcompile.sh` script file:
`sudo ./kcompile.sh`
 This script creates a file called `errors-6.19.9-moker` where it outputs the compilation messages.
5. The last step is to reboot the machine:
`sudo reboot`

7 Booting Linux kernel 6.19.9-moker version

So this is the last step, booting the "My Own Kernel" (moker) Linux kernel version.

1. During the system boot, when it presents the grub menu:
 - (a) Select "Advanced options for Ubuntu" and press the "Enter" key;
 - (b) Choose "Ubuntu, with Linux 6.19.9-moker" and press the "Enter" key.





2. Finally, we get our system.

